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(71)**Name of Applicant :**
1)RAJALAKSHMI ENGINEERING COLLEGE (AUTONOMOUS)
Address of Applicant :RAJALAKSHMI NAGAR, THANDALAM, CHENNAI, TAMIL NADU, INDIA Tamil Nadu India
(72)**Name of Inventor :**
1)DR.SURESH KUMAR S
2)Dr Sudalaimuthu T

(57) Abstract :
AGENTIC AI SYSTEM AND METHOD FOR SUSTAINABLE AGRICULTURE An agentic agricultural system (100) and method (200) are disclosed for optimising a cultivation environment. A perception subsystem (102) obtains environmental parameters of the environment. An agent orchestrator (104) determines a farm state and selects a field action by executing a closed perceive�reason�plan�act�learn loop, a reasoning module (106) generating a crop-suitability prediction, a pest determination, and an irrigation determination therefrom. An_actuation subsystem (108) executes the selected field action, comprising a control of a drip-irrigation valve, a dispatch of an unmanned aerial vehicle, or a control of a dosing pump, and an oversight module (110) refers a field action exceeding a consequence threshold to a human operator for approval. The agent orchestrator (104) adapts subsequent selections in dependence upon observed outcomes, whereby corrective action is carried out autonomously and a consumption of water and agricultural chemicals is reduced. Fig. 3

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